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Review article

# Brain systems at the intersection of chronic pain and self-regulation

Marianne C. Reddan, Tor D. Wager*

*University of Colorado, Boulder, United States*

ABSTRACT

Chronic pain is a multidimensional experience with cognitive, affective, and somatosensory components that can be modified by expectations and learning. Individual differences in cognitive and affective processing, as well as contextual aspects of the pain experience, render chronic pain an inherently personal experience. Such individual differences are supported by the heterogeneity of brain representations within and across chronic pain pathologies. In this review, we discuss the complexity of brain representations of pain, and, with respect to this complexity, identify common elements of network-level disruptions in chronic pain. Specifically, we identify prefrontal-limbic circuitry and the default mode network as key elements of functional disruption. We then discuss how these disrupted circuits can be targeted through self-regulation and related cognitive strategies to alleviate chronic pain. We conclude with a proposal for how to develop personalized multivariate models of pain representation in the brain and target them with real-time neurofeedback, so that patients can explore and practice self-regulatory techniques with maximal efficiency.

## 1. Introduction

Pain is a multidimensional experience, often described in terms of sensory-discriminative, cognitive-evaluative, and affective-motivational components [1,2]. These three categories encompass a variety of processes, including nociception [3], attention [4,5], learning [6], memory [7,8], expectation [9,10], personality [11], culture [12–14], socioeconomic conditions [15–17], and more. When peripheral nociceptive information enters the cortex, the signal becomes integrated with complex multidimensional information and is modified via descending projections [18–21]. Through this mechanism, the pain experience is not only heterogeneous, but personalized. Therefore, to understand pain, we must seek to understand the person in pain [22,23].

Broadening our conceptualization of pain to include dimensions outside of primary nociception and sensation complicates the task of neurobiological research, but it also expands the landscape of treatment possibilities. That is, when we reconceptualize pain as something *constructed in* the brain, rather than something that *happens to* the brain, we allow for the development and testing of noninvasive and non-pharmacological interventions, such as self-regulation—a family of cognitive strategies for managing thoughts, emotions, and actions that affects the brain's activity and neurochemistry and alters neuroplasticity over the course of time ([24–26]; Mischel, Cantor, & Feldman, 1996).

The exploration of nonpharmacological treatments for chronic pain is important. Opioids, though effective in the short term, may not provide long-term pain relief [27–29]. Furthermore, opioids are prone to abuse [30,31]. Aberrant addiction-like behaviors develop in approximately 24% of cases of chronic back pain patients prescribed opioids [32,33]. There is also a growing epidemic of opioid abuse and opioid-related death in Western countries [34,35], which can sometimes be localized to neighborhoods surrounding opioid-prescribing physicians [36]. Therefore, it is critical for scientists to prioritize investigations of non-opioid, as they are not only limited in their effectiveness, but potentially deadly.

Compared to other treatments, opioids are not significantly advantageous. Their efficacy for short-term pain management [37] is comparable to both cyclobenzaprine (a 5-HT2A antagonist, pro-adrenargic muscle relaxant [38];) and antidepressants [39]. Furthermore, nonpharmacological treatments of chronic pain, such as mindfulness [40], hypnosis [41,42], yoga [43], and combinations of psychological and physical treatment programs (biopsychosocial multidisciplinary rehabilitation [44];) produce pain-reducing effects comparable to that of opioids (Fig. 1). Self-regulatory treatments for chronic pain are therefore, a promising, albeit desirable, alternative to pharmacologic agents and are worthy of further research.

In this paper, we briefly discuss the heterogeneity of pain representations in the brain and identify several loci whose activity can be disrupted to alleviate chronic pain conditions through self-regulatory techniques. Specifically, we discuss (1) prefrontal-limbic circuitry and (2) resting state networks, which are altered during the chronification of pain. We propose that prefrontal-limbic circuitry alters the way pain is *valued* in individuals, while changes to default mode network (DMN) morphometry and connectivity may alter self-referential processes related to the pain experience. These systems can be

* Corresponding author at: Department of Psychology and Neuroscience, University of Colorado, 345 UCB, Boulder, CO, 80309, United States.
*E-mail address:* tor.wager@colorado.edu (T.D. Wager).

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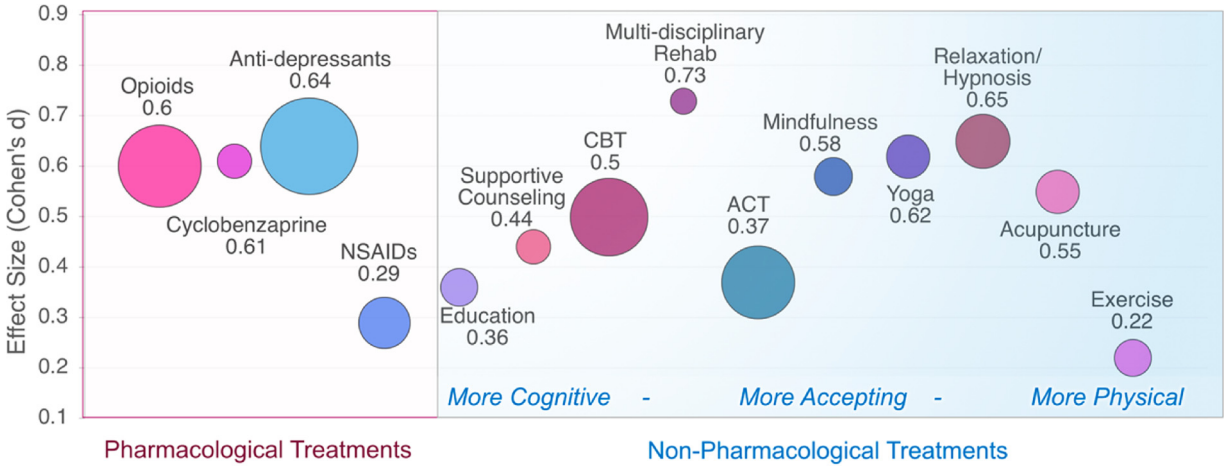



**Fig. 1.** Effect Sizes of Chronic Pain Treatments on Pain Reports. Several meta-analyses of controlled studies for different treatments of various types of chronic pain were selected. The effect size (Cohen's *d*), of the treatment effect is plotted here. Sizes of the bubbles indicate the number of studies included in the meta-analysis, therefore, larger bubbles are likely to be more reliable than smaller bubbles. Treatments are arranged from left to right where the left box, are pharmacological treatments (effect sizes relative to placebo), while the right box, are non-pharmacological treatments. Non-pharmacological treatments refer to self-regulation techniques that include cognitive-based, acceptance-based, and physical activity-based strategies in varying degrees. The largest effect size ($d = 0.73$) in this comparison was for a multi-disciplinary rehabilitation program for chronic lower back pain patients compared to a usual care control. The multidisciplinary rehabilitation in this meta-analysis was defined according to the biopsychosocial model of pain; a study was eligible for inclusion if the multidisciplinary intervention involved a physical component and a psychological (i.e., CBT or ACT) or a social/work component [44]. Please see Supplementary Table 1 for more details on the meta-analyses included. Further investigation is needed to better compare the durability of these effects in relation to disease progression.

targeted by self-regulatory strategies that alter evaluative (i.e., re-appraisal-based strategies) and self-referential processes (i.e., acceptance-based strategies), respectfully.

### *1.1. The heterogeneity of pain representations in the brain*

The experience of chronic pain is different from that of acute pain. Critically, it is *appraised* differently; its perceived causes are understood differently, and it connotes different implications for future wellbeing. Acute pain is often appraised as self-limiting, with an identifiable cause and expected resolution. Conversely, chronic pain is appraised as something without a known end that is dissociated from its initial driver and a part of daily life [45].

Likewise, different types of chronic pain are experienced and represented in unique ways. For example, chronic back pain disrupts activity in the mPFC more than post-herpetic neuralgia or knee osteoarthritis [46]. In post-herpetic neuralgia, the most prominent correlates of spontaneous pain are in the amygdala and accumbens, while knee osteoarthritis has greater correlations in the orbitofrontal cortex. Insula activity is central to brain changes associated with fibromylgia [47–50]. Migraine episodes evoke a distributed network of activity with a significant emphasis on subcortical circuitry— signal intensity in the red nucleus and substantia nigra as related to visually-triggered onset [51], and the pons as related to spontaneous migraine experiencing [52], migraine severity [53], and migraine-related changes in morphology [54]. In phantom pain, pain may be best explained by persistent local cortical representations in S1 [55] or maladaptive cortical reorganization [56]. To summarize, brain representations across chronic pain pathologies are unique, though they may implicate common circuitry that overlaps with acute pain at a broad level (Fig. 2).

For at least some types of pain—and chronic low back pain in particular—the transition from acute to chronic pain is thought to be supported by a shift in pain representation from nociceptive to 'emotional' circuits, that is, the limbic system and prefrontal cortex [57]. This is supported by a pattern of dissociation between brain systems involved in immediate evoked pain experience and those that seem to track pain chronification. For example, the Neurologic Pain Signature (NPS; [58]), a brain pattern predictive of evoked experimental pain, includes a set of brain regions that overlap with those often considered to be in the 'pain matrix' by virtue of their responses to noxious stimuli and correlations with pain ratings ([59–61,1],). Like the 'pain matrix,' in the NPS, activity in the anterior cingulate cortex (ACC), insula, secondary somatosensory cortex (S2), and thalamus is related to a greater pain experiencing, while increased activity in the ventromedial prefrontal cortex (vmPFC) and precuneus is related to *less* pain experiencing. Indeed, the regions that predict evoked pain as part of the NPS are not the regions that correlate with the intensity of spontaneous fluctuations in chronic back pain [46] nor does activity in these regions predict back pain chronification [62]. The region most heavily implicated in pain chronification is the vmPFC, which changes from being anti-correlated with pain in healthy participants [63] to being positively associated with pain in several patient groups [48–50].

One interpretation of this evidence is that there is a shift in the drivers of pain from systems that encode the immediate, sensory-discriminative aspects of pain experience to affective-motivational information relevant for pain avoidance, which requires linking actions and contexts with pain via associative learning mechanisms [6,64–66]. This is generally adaptive, as actions associated with pain should be avoided in order to prevent harm. However, avoidance can become maladaptive if a wide variety of actions and contexts become linked with pain—particularly if those actions are not harmful and the pain-avoidance system has over-generalized [67].

For example, pain from an injury may be exacerbated by long periods of standing. If a patient is required to stand for long periods of time at work, they may learn to associate their work environment with pain. Indeed, conditioned increases in muscle tension are acquired more rapidly in subacute pain patients at high risk for chronification [68]. That is, those at a higher risk of pain chronification demonstrate quicker associative learning between pain contexts and nociceptive responses. Furthermore, if the patient's community engages in movement-related hobbies, the patient may learn to expect pain during social events. This may promote social withdrawal and feelings of alienation [69,70]. One patient's testimony of their experience living with chronic pain elucidates this point:

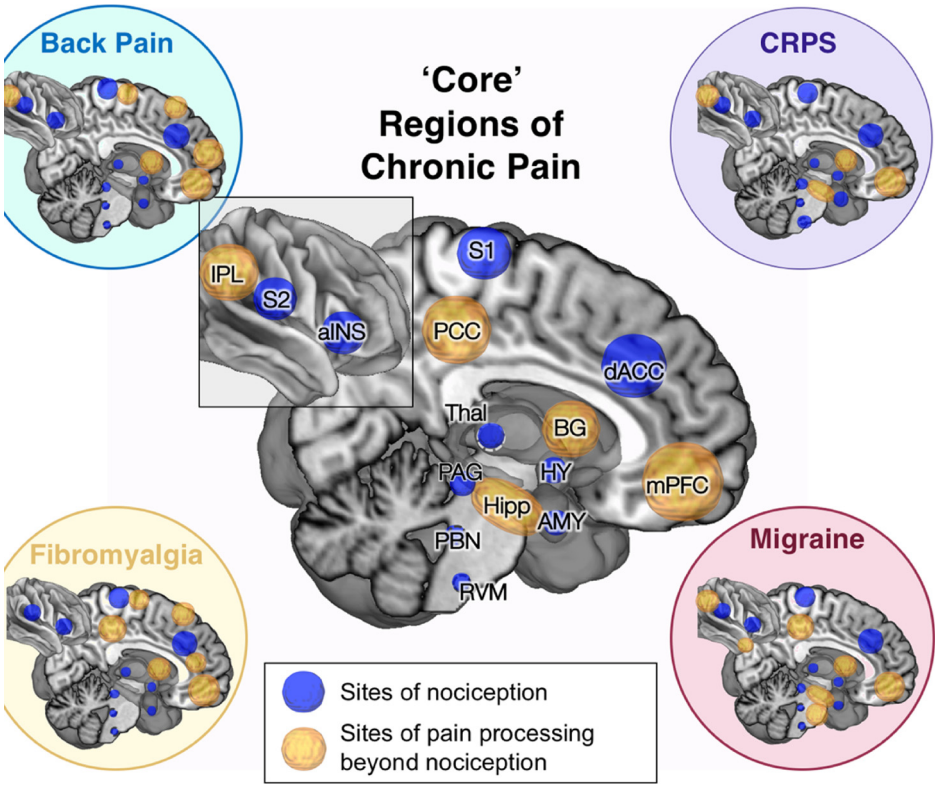



**Fig. 2.** Chronic pain brain representations are heterogeneous but may share a 'core' network. *Top Left.* The brain regions implicated in the experience of chronic back pain ([46], [205], [134], [206]; [208]; Loggia et al., 2015). *Top Right.* Brain regions implicated in pain experiencing in CRPS patients relative to controls ([204]; [220] [62,211];). *Bottom Left*. Brain regions implicated in fibromyalgia pain experiencing relative to controls ([50]; [48] [219];). *Bottom Right*. Brain regions implicated in the experience of migraine headaches in chronic migraine patients relative to baseline. ([216]; 2012; [53]; Grazzi et al., 2010; Welch et al., 2001; [51]; Schwedt & Dodick, 2009). *Center*. 'Core' Regions of Chronic Pain. Despite the heterogeneity in the brain representations of different chronic pain pathologies, some regions are consistently implicated across diseases. These include, sites of nociception (dark grey) such as the thalamus (Thal), dACC, SI, SII, and the anterior insula (aINS), as well as sites of pain-related pain processing beyond nociception (light grey) such as the hippocampus (Hipp), medial prefrontal cortex (mPFC), basal ganglia (BG), posterior cingulate cortex (PCC), and inferior parietal lobule (IPL). Both functional and morphological studies were surveyed to make this diagram.

**Table 1**
Functional Taxonomy of Pain Self-Regulation Strategies.

| Category | Description | Related clinical treatments |
|---|---|---|
| Reappraisal-based strategies | Focused attention to sensations, conscious modification of the meaning of sensation | Cognitive Behavioral Therapy (CBT) |
| Acceptance-based strategies | Focused attention to sensations without appraisal | Mindfulness meditation<br>Yoga<br>Acceptance and Commitment Therapy (ACT) |
| Attention-based strategies | Removal of attention from bodily sensations | Distraction, working memory tasks, engagement with external task/ stimuli |

> "In my 10+ years of chronic pain I've done many regrettable things, but the thing I regret is something I had no clue I was doing. I began pushing everyone away so that they didn't have to deal with me or so that I could just quietly decay in my bed without bother. It starts small, you begin missing more and more things and then people stop trying to invite you because they think you'll say no. I wish I was stronger is the only phrase I want to tell my best friend and how I wish I could have been at your wedding to see you happy." (Anonymous user on the forum: www.reddit.com/r/chronicpain).

Depressed mood, negative pain beliefs, and early beliefs that the pain may be permanent are also predictive of the chronification of neck and back pain [71]. In this way, various aspects of a person's life may become entangled with their pain experience, and a widening variety of activities that could promote recovery and wellbeing become threatening and are avoided. That is, chronic pain, unlike acute pain, is conceived as something inseparable from one's sense of *self*. This pathological interweaving of pain and self is therefore a promising target for treatment through self-regulatory techniques.

## 2. Pain self-regulation

Self-regulation encompasses a wide variety of strategies for deploying conscious thought to shape attention, emotion, decision-making, and behavior [24–26]; Mischel, Cantor, & Feldman, 1996). It has been proven to be effective for the attenuation of both experimental pain [72] and chronic pain [73–77]. Furthermore, it is a feasible option for long-term treatment because it is a free, learnable skill set that can be improved with practice [25,78,79]. When practiced effectively, self-regulatory strategies can reduce reliance on drugs, surgery, and other high-cost and high-risk treatments [80–82]. Self-regulatory interventions that induce learning or neuroplasticity in brain regions that undergo reorganization or functional connectivity changes during the chronification of pain may be able to slow down, mitigate, or reverse the effects of chronic pain pathologies [83]. Here we identify three categories of self-regulatory strategies with the potential to alter the neural substrate underlying chronic pain: (1) reappraisal-based [84], (2) acceptance-based [85], and (3) attention-based [86] strategies (Table 1).

Pain ***reappraisal***, a type of cognitive reframing of the context and meaning of pain, is a part of a family of processes studied extensively in relation to emotion [87,88]. Reappraisal involves changing an emotional experience by reinterpreting the meaning of the emotion-eliciting stimulus. During pain reappraisal, one's perceptions of pain and pain reactivity are consciously modified by changing the meaning of the pain (i.e., reinterpreting pain as non-dangerous, reminding oneself that it is time-limited and under control, affirming that 'suffering builds character,' and more; [89–91]; Sturgeon, Kao, Hah, and Mackey, 2014).

Through reappraisal, the value of pain is consciously altered to either down-weight its negative components or up-weight its positive components. It may remain salient, but its meaning is intentionally modified.

Pain ***acceptance*** is the process of learning to acknowledge pain's existence as neither positive nor negative. It requires one to stop resisting pain, but instead, to learn to live with it without judgment, and to notice and reduce fear and avoidance of pain. Acceptance-based therapies have been shown to decrease both pain experiencing and psychological distress [92–95]. Through acceptance, the value of the pain is neutralized; it is no longer salient though it is still consciously accessible.

***Attentional control*** focuses on directing attention away from the pain experience [96,97]. Through attentional control, the actual value of the pain is not changed, instead its accessibility is changed. The redirection of attention can be very effective for evoked pain [98–100]; however, it may not be sustainable when pain is chronic [101,102]. Furthermore, there is evidence that distraction from chronic pain during pain-inducing activity has a paradoxical effect of increasing pain immediately after the task [103]. Therefore, we focus here on reappraisal- and acceptance-based strategies.

### 2.1. Reappraisal-based strategies

#### 2.1.1. Behavioral effectiveness

A popular treatment that often utilizes reappraisal is cognitive-behavioral therapy (CBT), a psychotherapy that seeks to alleviate dysfunction through the modification of emotions, behaviors, and thoughts [104,105]. CBT is reportedly efficacious for the treatment of pain and stress-related disorders [77,106–109] with treatment effects greater than some drugs and other education-based therapies (see Fig. 1). For example, CBT has been shown to provide pain relief from headaches [110,111], and its protective effects last through six- to nine-week follow up periods [112,113]. CBT promotes general improvement in a variety of health-related outcomes in chronic pain patients, including pain experiencing, mood, physical activity, coping ability, and sociality [106]. CBT also reduces the economic as well as societal impacts of chronic pain; in an investigation of 409 chronic low back pain patients, patients who had CBT in combination with usual care in a 3-week inpatient rehabilitation program had 5.4% fewer work absences six months later than those who received usual care alone [114].

Reappraisal therapies teach techniques to help endure or mitigate the current pain experience. They also teach ways in which a patient can alter environmental, cognitive, and affective factors that may trigger or exacerbate pain. In this way, CBT is a method of 'unlearning' pain [115–117], which can change one's expectations, beliefs, and cognitions about one's pain. A cognitive component related to persistence and intensity of pain which can be targeted by CBT is catastrophizing [77,118,119]. Pain catastrophizing is a key feature of the Fear-Avoidance Model of Pain, which describes two paths for patients after an injury [120–123]. Patients who develop chronic pain after an injury may begin to fear pain, associating negative affect, vigilance towards potential pain, and negative beliefs with their experience of pain. For example, an item on the Fear-Avoidance Components Scale is "*I believe that my pain will keep getting worse until I won't be able to function at all.*" Such beliefs may lead to hypervigilance and generalized avoidance of physical and social activity — behaviors that may worsen musculoskeletal and central nervous system drivers of pain (i.e., disuse may increase disability), and also create socio-emotional distress (i.e., social withdrawal and feelings of isolation may lead to depression). In this model, if patients "confront" their pain experience, they are more likely to recover. CBT is a tool patients can use to effectively "confront" their pain by understanding this vicious cycle of pain and fear, and changing their beliefs about their pain [124–126]. Patients who interpret their pain as non-threatening are less likely to catastrophize or engage in pain-related fear-avoidance behaviors, and demonstrate quicker recovery times [120].

#### 2.1.2. Brain targets of reappraisal strategies

The basal ganglia (BG) is a crucial structure for both valuation (for a review [127]) and multisensory integration [128,129] implicated in both pain experiencing [130] and emotion reappraisal [131]. There is evidence that thalamo-cortico-BG loops integrate sensorimotor, affective, and cognitive information into the pain experience [130,132,133], indicating that plasticity in prefrontal-limbic circuitry changes the way nociceptive signals are appraised, evaluated, and associated with other types of processing. Indeed, pain chronification in humans [134] and rats (Metz, Yau, Centeno, Apkarian, & Martina, 2009; [135]) induces structural changes in the PFC and nucleus accumbens (NAc) that promote pain avoidance and depression-like behavior. In subacute back pain patients, the degree of functional connectivity between the mPFC and the NAc predicts transition to chronic pain [62], and the strength of mPFC-NAc connectivity is correlated with a "shift" in the neural representation of back pain from primarily nociceptive to emotion processing brain regions [57].

In a rat model of neuropathic pain [136], peripheral nerve injury selectively sensitized an indirect pathway between the NAc shell and spiny projection neurons (which constitute ~95% of cells in the mammalian striatum [137]. This experimentally induced tactile pain sensitization was reversed when spiny projection neurons were inhibited and exacerbated when spiny projection neurons were excited. The authors interpret this finding as evidence that spiny projection neurons of the striatum not only participate in the central representation of pain, but gate activity in ascending nociceptive pathways [136]. Drawing from this evidence, the PFC and NAc have been identified as part of a 'gatekeeping' system, which calculates a stimulus' relevance (saliency) and affective value, and modulates information flow in the descending modulatory pain system [138,139]. Self-regulatory techniques which target evaluative processes may alter the pain experience through modulation of this circuitry.

In studies of emotion regulation, cognitive reappraisal increases activity in brain regions associated with the processing of affective meaning: the mPFC, anterior temporal cortices, and the ventral striatum [140]. It also decreases activity in limbic regions, particularly the amygdala [141–143,88], which may be associated with individual differences in rumination [144]. Using a mediation analysis, Wager et al. [131] found two independent pathways for cognitive reappraisal via the ventrolateral PFC (vlPFC): (1) a pathway between the vlPFC and amygdala which predicted greater negative emotion reports, that is, failures of reappraisal, and (2) a pathway between the vlPFC and NAc (and/or surrounding ventral striatum) that predicted greater positive emotion reports, that is, reappraisal success. Focusing on NAc as a seed, Wager et al. [131] identified an additional pathway from the vmPFC to NAc supporting reappraisal success. Similarly, self-regulation of evoked heat pain can be explained by a pathway connecting the vmPFC and the NAc [91]. In animals, stimulation of the PFC can inhibit both sensory and affective pain signals [145] via descending projections to the striatum [146,147], amygdala [148], brainstem, and spinal cord [149]. Through this mechanism, activation of the PFC during CBT may alter the valuation of the pain experience via its projections to limbic circuitry, and therefore, facilitate the reappraisal of pain beliefs.

### 2.2. Acceptance-based strategies

#### 2.2.1. Behavioral effectiveness

Acceptance-based strategies for the regulation of pain and negative affect are rising in popularity, and hold promise for long-term pain management [150–152]. One favored form of acceptance-based therapy is mindfulness meditation. Mindfulness meditation is the act of focusing attention on bodily sensations on a moment-by-moment basis, and removing judgement attached to these sensations [153]. Thus, to be mindful of pain, one focuses on the experience of it unfolding

through time but removes appraisals of what the experience means (i.e. a signifier of tissue damage). The psychological processes involved in mindfulness would suggest that it does not directly attenuate nociceptive signals or pain sensations, but acts by manipulating non-nociceptive qualities of pain. Indeed, experienced meditators report similar levels of pain intensity as non-meditators, but report reduced levels of pain unpleasantness [154–156]. Compared to rest, however, the act of meditation can reduce both pain unpleasantness and intensity ratings to acute noxious stimulations [157].

Mindfulness interventions have been effective for patients with fibromyalgia [158–160], musculoskeletal pain [161,162], rheumatoid arthritis [163,164], and headache (Nash-Mc Feron 2005); however these effects are often estimated by comparing the treatment group to no-treatment wait-listed controls, and thus include a wider variety of 'non-specific' effects of engagement in treatment and positive expectancy than most drug trials [165]. Still, mindfulness is reported to have a significant impact on distress, depression, and other affective dimensions of the pain experience [166], and in some cases it is more effective for long-term pain management than CBT or usual care [152]. Mindfulness may work by reducing distracting and ruminative thoughts [167]. Other potential psychological mechanisms are meta-awareness, the bringing to consciousness feelings of fear and anxiety surrounding pain, and de-identification with both emotions and pain (i.e., "Pain is an experience passing through me temporarily; it is not me and does not define me."). Increasing awareness of the 'vicious cycle' of fear and avoidance may in turn help increase engagement in activities and break the cycle. Indeed, Schütze et al. [168] found that patients who were practiced in mindfulness had lower rates of pain catastrophizing, as well as lower pain intensity, negative affect, and functional disability.

#### 2.2.2. Brain targets of Acceptance-based Strategies

Experienced meditators can attenuate elements of pain by deactivating areas of the brain that process the meaning of pain (i.e., PFC), while leaving the sensory experience of it intact or even enhanced [169,170]. Strikingly, several studies have failed to find any decreases in activity in correlates of evoked pain (e.g., posterior insula, SII, mid-insula, medial thalamus), and instead find increases in somatosensory processing regions including SII [157,170–172]. Experienced mindfulness meditators also demonstrate differences in default mode network (DMN) activity [173]. Core regions of the DMN include the mPFC, PCC, angular gyrus, lateral temporal cortex, hippocampus, parahippocampus. and inferior parietal lobule (IPL [174];). The DMN is active when an individual is engaged in internally directed thought—thinking about themselves or others, remembering the past, or imagining future events [174,175]. It becomes deactivated when an individual's attention is directed toward external tasks [176]. Accumulating evidence suggests that the DMN is critical for self-referential processing [177–180].

The DMN has been shown to interact with the PAG to support analgesia when one "mind-wanders away from pain" in healthy adults experiencing *acute* pain, a finding which suggests that one's ability to switch between resting state networks may underlie individual differences in pain disengagement [181]. However, for many chronic pain patients, "mind-wandering away from pain" is not viable long-term, and the more self-referential processing they engage in during their pain experience, the more their disease progresses [182,183]. Indeed, increased DMN activity in chronic pain is correlated with increased rumination and pain-related depression [181,184,185]. Meditation, therefore, may help to reduce the frequency of mind-wandering and self-referential processing which may help to dissociate the experience of pain from one's concept of self.

Various chronic pain pathologies are marked by abnormalities of resting-state functional connectivity networks including the DMN, salience network, and sensorimotor network (Table 2). Many of the regions that encode immediate pain experience are included in the 'salience' and 'sensorimotor' networks. A putative function of the salience network (or 'ventral attention' network) is to identify the most relevant internal and external stimuli [186]. It is comprised of the ventrolateral PFC, anterior insula, ACC [187], as well as subcortical structures including the amygdala and BG [188,189]. The 'salience network' may influence activation in the DMN by aiding in the 'switching' between intrinsic and extrinsic loci of attention [190]. Greater connectivity between the DMN and the salience network may be related to increased disease activity and clinical pain in chronic pain patients [191]. The 'sensorimotor' or 'somatomotor' network primarily integrates information about bodily action and sensation, and involves the primary somatosensory and motor cortices [187,192]. Connectivity between the DMN and the sensorimotor network may be related to physical functioning in chronic pain patients, where increased connectivity between M1 and the PCC predicts functional impairment [191].

**Table 2**
Reported evidence of changes to resting-state networks across chronic pain pathologies relative to healthy controls.

| Pathology | Default Mode Network | Salience Network | Sensorimotor Network |
|---|---|---|---|
| Chronic back pain | altered[1,2,3,4,5,22] | altered[24] | altered[5,24] |
| Arthritis | altered[4,6,7,8] | altered[6,7] | altered[20] |
| Fibromyalgia | altered[9,10,11] | altered[9,6] | altered[17] |
| CRPS | altered[12,4] | | altered[12] |
| Chronic pelvic pain | altered[13] | | altered[18] |
| Migraine | altered[14] | altered[14] | |
| Diabetic neuropathic pain | | altered[19] | |
| Idiopathic temporomandibular disorder | altered[15] | | |
| Chronic orofacial pain | altered[21] | | |
| Burning Mouth | altered[23] | | |
| Phantom pain | altered[16] | | altered[16] |

[1] [205], [2] [222], [3]Ceko et al 2015, [4] [206], [5] [217], [6] [221], [7] [191], [8] [194], [9] [50], [10] [212], [11] [221], [12] [207], [13] [218], [14]Hubbar et al, 2014, [15] [184], [16] [215], [17] [214], [18] [213], [19] [209],[20] [210], [21]Alshelh et al., 2018, [22]Loggia et al, 2013, [23]Shariq et al, 2014, [24]Kolesar et al., 2017.
A list of all studies to date (to our knowledge) that report alterations to the 'default mode,' 'salience,' or 'sensorimotor' networks in patients with different types of chronic pain. Alterations are not reducible to increases or decreases in relation to pain across studies as there may be increases between two regions and decreases between two others within one network, so for brevity, this chart only indicates if any changes relative to neurotypicals were found. This chart includes functional as well as structural findings.

Normally, the DMN is anti-correlated with the 'salience network,' and has no strong relationship with the 'sensorimotor network' [176,193]. In chronic pain, DMN and the 'sensorimotor network' demonstrate 'hyperconnectivity,' while DMN and the 'salience network' reduce in their anti-correlation, or in some cases, become positively correlated [191,194]. Hemington et al. [194] propose that the experience of pain initially evokes the 'salience' and 'sensorimotor' networks, as one must attend to their pain and potentially take action to escape it. However, during the chronification of pain, as the pain persists and becomes a part of daily functioning, people will naturally engage in self-referential processing while enduring pain. Therefore, pain-related processing in the salience network may no longer serve to suppress DMN activity, and 'sensorimotor' network activity may become correlated with DMN as the person begins to integrate their experience of pain with their sense of self [195]. In this way, the pain signal becomes integrated with "non-nociceptive" self-referential processing streams. Acceptance-based self-regulation strategies may work to prevent or undo this pathological association between one's sense of self and one's experience of chronic pain by preserving or enhancing anti-correlation between nociceptive and self-referential processing systems.

Overall, research on the self-regulation of pain indicates that pain regulation techniques interact with pain signaling to promote relief. Attention-based techniques such as distraction engage the descending

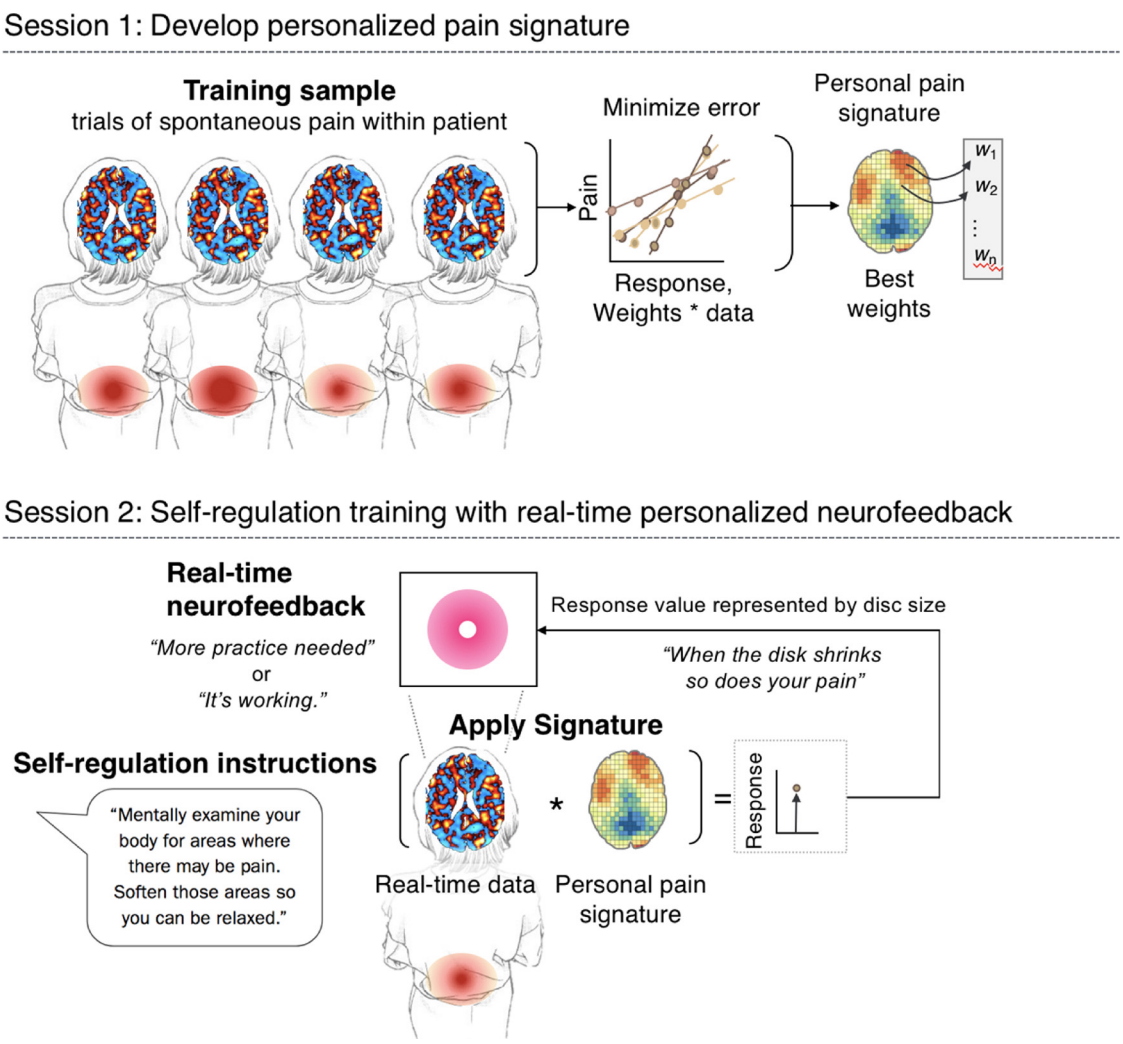



**Fig. 3.** Towards self-regulation training that targets personal experiences of pain. A proposed framework for studying the effects of self-regulation training on spontaneous chronic pain intensity. *Top Panel:* Whole brain multivariate fMRI patterns of brain activity can be recorded from a patient across may trials of spontaneous pain. Participant self-reports of the intensity of their pain can be used to train a personalized brain-based biomarker of chronic pain within that individual. *Bottom Panel:* A participant can practice various self-regulation strategies, such as mindfulness, while their brain activity is recorded. As they practice, real-time feedback on the expression of their personalized pain signature can be displayed in various formats, such as a disc that shrinks as 'pain,' determined by their brain activity, decreases.

pain modulatory system; reappraisal can influence pain by changing its valuation, or what it means to the self; and acceptance-based strategies such as mindfulness can reduce pain either by diminishing or manipulating pain-related self-referential processes. Together, this evidence supports the idea that pain can be modified in meaningful ways by non-somatic interventions, that is, interventions that do not treat bodily tissue, but which target the *experience* of pain.

### 2.3. Towards personalized models of pain experience

The earlier discussion of the heterogeneity of pain experiencing suggests that no two chronic pain conditions are the same, and that even within pathologies, pain is a personal experience. Therefore, despite commonalities in network dysfunction across a wide range of chronic pain disorders, the neural representation supporting any individual's experience of pain will necessarily reflect that person's unique integration of self-relevant information into their own pain experience. This is because the volume and content of information processed in these cortical regions is diverse and expansive, so when pain signaling is integrated into distributed cortical processing streams, it becomes much more complex than the peripheral input from which it originated. It is therefore possible that the pain experience itself *emerges* from the complex interactions occurring throughout these recurrent processing streams [196–198] and therefore, depends upon diverse neurobiological processes in the context of one's daily life.

To understand the complexity of the pain experience, we propose that future research be devoted to the development personalized neuroimaging models of pain experiencing. Using pattern-based multivariate modelling of whole brain activity during spontaneous pain experiencing in chronic pain patients, researchers can develop biomarkers sensitive and specific to pain within individual patients [199]. An individual's brain-based biomarker could then be applied to their subsequent brain activity in order to predict the level of pain they are experiencing at any given moment (see Fig. 3 for an illustration). If this is applied in real-time, a patient could receive immediate feedback on their neural representation of pain. This is most useful when a patient is learning how to self-regulate their pain experience. For example, a patient who is interested in mindfulness but is having trouble learning which strategies work best for them, could try different strategies in the fMRI scanner while receiving feedback on their ability to downregulate their personal pain signature (similar frameworks have been attempted for fear-regulation see [200]). In this way, a patient could learn through

trial and error what self-regulation strategies work best for them.

## 3. Conclusion

In this paper we ask: Can self-regulation impact the biological mechanisms that generate and perpetuate chronic pain? To answer this question, we first accepted and defended two premises: (1) that pain is a multidimensional experience, instantiated by emotional and cognitive systems in conjunction with primary nociception, and (2) that self-regulation is a cognitive process that affects affective-motivational processing and behavior. What may link these two processes, so that they can effectively influence one another, is their involvement of emotional and cognitive systems. If self-regulation affects emotional and cognitive brain mechanisms that directly contribute to chronic pain it is likely to be neurobiologically meaningful for patient recovery. Therefore, we sought to establish where in the brain these two experiences, that of chronic pain and that of self-regulation, interact neurobiologically. We propose that prefrontal-striatal circuitry and DMN functioning are critical entry points by which self-regulation can directly influence the chronification of pain.

Self-regulation is a cost-effective and scalable treatment available immediately to large populations, via clinics, educational programs, and even smartphone apps [201]. Furthermore, self-regulation works safely in conjunction with currently prescribed drug regimens, allowing any patient to simply and effectively supplement their ongoing treatment. Finally, self-regulation is a continuous and preventative measure for healthy populations to minimize the risk of developing chronic pain in the first place. Pain self-regulation therapies holds great promise for people suffering from chronic pain. Through continued study of the neurological underpinnings of self-regulation and how it impacts learning and neural plasticity, we will be able to optimize and personalize self-regulation training, understand how it can augment drug treatments, and accelerate its integration into therapeutic treatFment programs.

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